

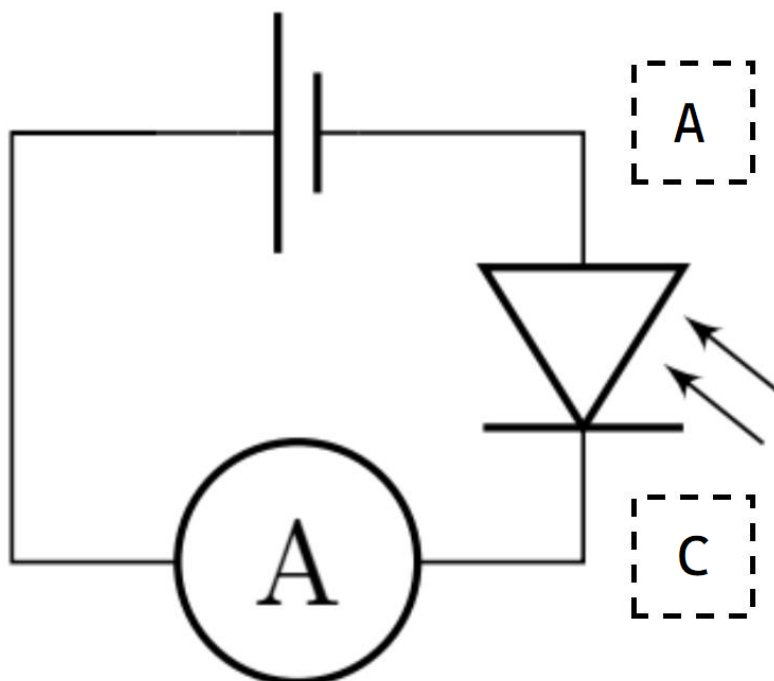
Casein titration – Solution

X25 (2025) — English translation

A1

0.10 pts

Using the photo, indicate on the diagram on your answer sheet which photodiode contacts each wire A, C corresponds to.



A2

0.50 pts

Position 5 cuvettes with 2.8 mL water so that the laser beam passes through them and falls on their walls normally. By successively adding 0.1 mL milk solution to each of the cuvettes, remove the dependence of the transmittance α on the number of cuvettes N in which the milk solution is added. Sketch the optical design used for $N = 3$.

Background radiation $I_{\text{back}} = 1.6$ mA. The transmittance is then given by

$$\alpha = \frac{I - I_{\text{back}}}{I_0 - I_{\text{back}}}$$

When making measurements in this part, it is important to take into account reflections at the interfaces, i.e. to study the light intensity for a given N , place N cuvettes with a milk solution and $5 - N$ cuvettes with clean water.

Answer: \textbf{Answer: } $I, \mu\text{A}$ N α 118.10 0 1.000 58.10 1 0.483 32.30 2 0.260 16.10 3 0.121 9.60 4 0.065 6.30 5 0.036

A3**0.50 pts**

Position one cuvette with 2.8 mL water so that the laser beam passes through its walls normally. Remove the dependence of the transmittance coefficient α on the volume ΔV of the added milk solution in the range $\Delta V \leq 1$ mL.

Answer: \textbf{Answer: } ΔV , mL I , μA α 0 262 1.000 0.1 139.7 0.530 0.2 81.6 0.307 0.3 48.9 0.182 0.4 29.4 0.107 0.5 18.6 0.065 0.6 13.6 0.046 0.7 9.8 0.031 0.8 7.0 0.021 0.9 5.6 0.015

A4**0.30 pts**

Let a parallel beam of light fall on a layer of milk solution with a thickness of L and a micelle concentration of n . Express the transmittance α in terms of L , n and the effective scattering area δS of the micelles.

The volume layer $dV = S \cdot dL$ contains $dN = dV \cdot n$ micelles, each of which absorbs light falling on it. Then since the intensity of the absorbed light is proportional to the total absorption area in this layer:

$$dI = -I \cdot \frac{dN \delta S}{S} = -I \cdot \delta S \cdot dL \cdot n$$

Integrating this equation, we obtain

$$I = I_0 e^{-Ln \delta S}$$

Answer: \textbf{Answer: }

$$\alpha = e^{-Ln \delta S}$$

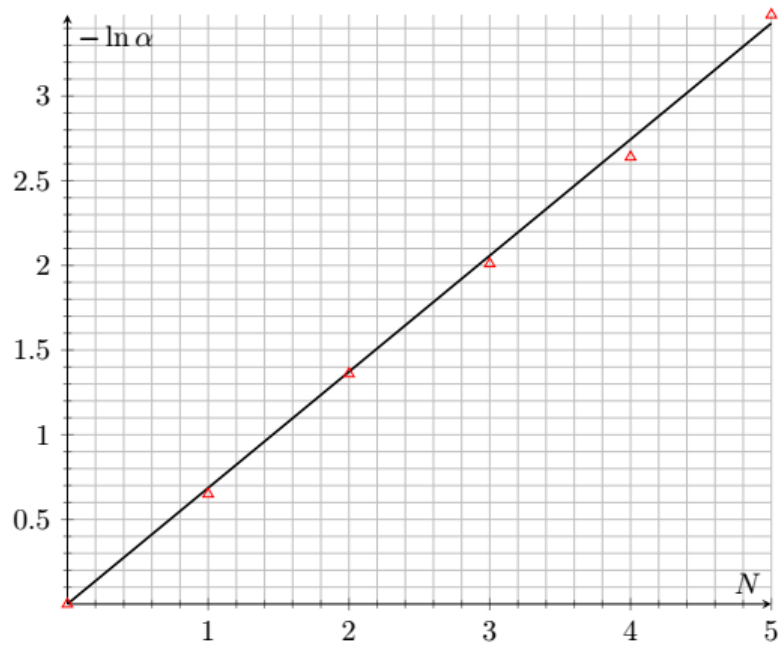
A5**0.70 pts**

Linearize the dependence of α on N from A1 and plot it. Determine the parameters of the line.

In the experiment under consideration, the length of the absorbing layer $L = L_0 \cdot N$ changes. The answer to the previous paragraph implies the linearization

$$\ln \alpha = -N \cdot L_0 n \delta S$$

The straight line passes through zero and has a slope coefficient $k_{A5} = 0.668$.



A6

0.80 pts

Linearize the dependence of α on ΔV from A2 and plot it. Find the parameters of the line.

In the experiment under consideration, the concentration of micelles changes. Let their concentration in the original milk solution be n_0 , then in the one we have considered

$$n = n_0 \frac{\Delta V}{V + \Delta V}$$

, then linearization:

$$\ln \alpha = -\frac{\Delta V}{2.8 \text{ mL} + \Delta V} \cdot L_0 n_0 \delta S.$$

The straight line passes through zero and has a slope coefficient $k_{\text{A6}} = 17.2$.